# Chapter 12 — Nuclear and Particle Physics

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## Q.1 — Multiple Choice Questions (with correct option and reason)

### 12.1. Which one of the following is a fundamental particle?

**(a)** proton **(b)** neutron
**(c)** electron **(d)** meson

**Answer: (c) electron** — Electron has no internal structure.

### 12.2. The first discovered anti-particle is:

**(a)** antiproton **(b)** anti-neutrino
**(c)** anti-photon **(d)** anti-electron

**Answer: (d) anti-electron** — The positron was discovered first.

### 12.3. Which pair of particles creates annihilation?

**(a)** proton-proton **(b)** proton-neutron
**(c)** neutron-photon **(d)** electron-positron

**Answer: (d) electron-positron** — Particle–antiparticle annihilation.

### 12.4. The strong nuclear force between two particles is mediated by:

**(a)** gluons **(b)** photon
**(c)** mesons **(d)** gravitons

**Answer: (a) gluons** — At quark level, gluons carry the strong force.

### 12.5. Which force interacts between two particles through photons?

**(a)** strong nuclear force **(b)** weak force
**(c)** electromagnetic force **(d)** gravitational force

**Answer: (c) electromagnetic force** — Photons carry the electromagnetic force.

### 12.6. When a neutron changes into a proton, we observe:

**(a)** $\beta^-$ decay **(b)** $\beta^+$ decay
**(c)** $\gamma$ decay **(d)** $\alpha$ decay

**Answer: (a) $\beta^-$ decay** — $\beta^-$ emits an electron and antineutrino.

### 12.7. Baryons are formed by combination of:

**(a)** 2 quarks **(b)** 3 quarks
**(c)** 4 quarks **(d)** a quark and an anti-quark

**Answer: (b) 3 quarks** — Baryons are three-quark states.

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**12.8. Which force has negligible effect between elementary particles?**

**(a)** strong nuclear force **(b)** weak force
**(c)** gravitational force **(d)** electromagnetic force

**Answer: (c) gravitational force** — Gravity is extremely weak at particle scale.

**12.9. Which particles are produced by strong interaction?**

**(a)** graviton **(b)** leptons
**(c)** hadrons **(d)** mesons only

**Answer: (c) hadrons** — Strong interaction produces hadrons.

**12.10. A strong nuclear force exists between the nucleons of:**

**(a)** p-p **(b)** n-n
**(c)** p-n **(d)** all of these

**Answer: (d) all of these** — It acts between all nucleon pairs.

**12.11. Which radiation/particle has the highest ionization power?**

**(a)** $\alpha$ **(b)** $\beta^+$
**(c)** $\beta$ **(d)** $\gamma$

**Answer: (a) α** — Alpha particles are heavy and doubly charged.

**12.12. Which radiation/particle has the highest penetrating power?**

**(a)** $\alpha$ **(b)** $\beta^+$
**(c)** $\beta^-$ **(d)** $\gamma$

**Answer: (d) γ** — Gamma rays are most penetrating.

**12.13. A change occurs in atomic number but mass number remains the same by decay of:**

**(a)** $\alpha$ **(b)** $\beta$
**(c)** $\gamma$ **(d)** $\alpha$ and $\gamma$

**Answer: (b) β** — Beta decay changes Z but not A.

**12.14. In a nucleus, a neutron changes into a proton; the mass number will:**

**(a)** decrease **(b)** increase
**(c)** remain the same **(d)** none of these

**Answer: (c) remain the same** — Total nucleons remain the same.

**12.15. The electroweak theory was introduced by:**

**(a)** Dirac **(b)** Einstein
**(c)** Anderson **(d)** Dr. Abdul Salam

**Answer: (d) Dr. Abdul Salam** — Salam contributed to electroweak unification.

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### 12.16. The asymmetry of matter and anti-matter is due to imbalance number of:

**(a)** hadron **(b)** lepton
**(c)** baryon **(d)** photons

**Answer: (c) baryon** — A small baryon excess survived.

### 12.17. Which particle is responsible for the mass of fundamental particles?

**(a)** quarks **(b)** anti-quark
**(c)** lepton **(d)** Higgs boson

**Answer: (d) Higgs boson** — The Higgs field gives mass.

### 12.18. A proton is composed of up and down quarks in the order:

**(a)** udd **(b)** udu
**(c)** uud **(d)** dud

**Answer: (c) uud** — Proton = uud.

### 12.19. The number of quarks that compose a neutron is:

**(a)** 2 **(b)** 3
**(c)** 4 **(d)** 5

**Answer: (b) 3** — Neutron = udd, three quarks.

## Q.2 — Short Answer Questions

### 12.1. What do different isotopes of an element have in common? How are they different?

Isotopes have the same atomic number Z (same protons) but different neutron numbers, so their mass numbers A differ.

### 12.2. Identify the element with 87 nucleons and 50 neutrons.

$Z = A - N = 87 - 50 = 37$, which is rubidium (Rb).

### 12.3. What are similarities and differences between strong nuclear force and electromagnetic force?

Both are fundamental forces, but the strong force acts between quarks/nucleons and is short-range and much stronger; the electromagnetic force acts between charges, is long-range and can attract or repel.

| Feature | Strong | Electromagnetic |
|---|---|---|
| Acts on | Quarks/nucleons | Charges |
| Range | Very short | Infinite |
| Relative strength | 1 | $\sim 10^{-2}$ |

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### 12.4. Fill in the missing particle or nucleus in the printed decay equations.

In beta-minus decay a neutron changes to a proton, emitting an electron and an antineutrino; in beta-plus decay a proton changes to a neutron, emitting a positron and a neutrino. The missing particles follow charge and nucleon-number conservation.

### 12.5. Why must a neutrino be released in positron emission?

It conserves lepton number, energy and momentum. Without it, the observed continuous beta spectrum and spin balance could not be explained.

### 12.6. Distinguish between fermions and bosons.

Fermions have half-integer spin and obey the Pauli exclusion principle, e.g. electrons and quarks. Bosons have integer spin and can occupy the same state, e.g. photons and gluons.

### 12.7. How does the strong force hold the nucleus?

It overcomes electrostatic repulsion between protons at short range, binding protons and neutrons together.

### 12.8. Can pair production occur for photons of energy 20 keV?

No. The minimum energy is 1.02 MeV, the rest energy of an electron–positron pair. 20 keV is far too small.

### 12.9. What is the difference between a beta particle and an electron?

A beta-minus particle is an electron, but it is created in the nucleus during decay, not an orbital electron.

### 12.10. How do a proton and neutron convert into each other?

In $\beta^-$ decay a neutron becomes a proton (d → u); in $\beta^+$ decay a proton becomes a neutron (u → d), with leptons emitted.

### 12.11. Why does beta decay have a continuous spectrum but alpha decay a discrete spectrum?

Beta decay shares energy among three bodies, so the beta particle can have a range of energies. Alpha decay is effectively two-body, giving discrete alpha energies.

### 12.12. Differentiate between hadrons and leptons with examples.

Hadrons feel the strong force and are made of quarks, e.g. protons and pions. Leptons do not feel the strong force, e.g. electrons and neutrinos.

### 12.13. Why can an electron–positron pair not decay into a single photon?

A single photon cannot conserve both energy and momentum in the centre-of-mass frame; two photons are needed.

### 12.14. State the role of the Higgs boson in generation of mass.

The Higgs field interacts with fundamental particles, giving them mass. The Higgs boson is the quantum excitation of this field.

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### 12.15. What are mesons? Give examples.

Mesons are hadrons made of a quark and an antiquark. Examples include pions and kaons.

## Constructed Response Questions

### 12.1 Is a meson a boson or fermion? Give reason.

A meson is a boson because it consists of a quark and an antiquark, giving integer total spin.

### 12.2 Why does an alpha emitter emit alpha particles instead of four separate nucleons?

The alpha particle is very tightly bound, so emitting it releases more energy and is more favourable than emitting four separate nucleons.

### 12.3 Which is more energetic, alpha decay or beta decay? Justify.

Alpha particles often have several MeV of discrete energy, while beta particles have a continuous spectrum up to a maximum. Typical alpha energies are larger than average beta energies.

### 12.4 A nucleus emits a gamma photon of 1.5 MeV. Calculate (i) frequency (ii) wavelength (iii) momentum.

$E = 1.5\ \text{MeV} = 2.4 \times 10^{-13}$ J. (i) $f = E/h \approx 3.6 \times 10^{20}$ Hz. (ii) $\lambda = c/f \approx 8.3 \times 10^{-13}$ m. (iii) $p = E/c \approx 8.0 \times 10^{-22}$ kg m $s^{-1}$.

### 12.5 Why does an alpha particle not make physical contact with a nucleus when headed directly toward it?

The positive alpha particle is repelled by the positive nucleus. It stops at the distance of closest approach where kinetic energy becomes electrostatic potential energy.

## Comprehensive / Long Questions

### 12.1 What is radioactivity? Compare properties and behaviour of three types of radiations.

Radioactivity is spontaneous nuclear decay. Alpha: helium nuclei, +2 charge, high ionization, low penetration. Beta: electrons/positrons, ±1 charge, moderate ionization and penetration. Gamma: photons, no charge, low ionization, high penetration.

| Radiation | Nature | Charge | Ionization | Penetration |
|---|---|---|---|---|
| α | He nucleus | +2 | Very high | Low |
| β | electron/positron | ±1 | Moderate | Moderate |
| γ | photon | 0 | Low | High |

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### 12.2 Elaborate beta-positive and beta-negative decay with examples.

$\beta^-$: $n \rightarrow p + e^- + \bar{\nu}$, e.g. neutron-rich nuclei. $\beta^+$: $p \rightarrow n + e^+ + \nu$, e.g. proton-rich nuclei. In both, A is unchanged and Z changes by ±1.

### 12.3 What is the difference between matter and antimatter? Why is the universe almost entirely matter?

Antimatter particles have the same mass but opposite charge. The early universe produced almost equal amounts, but a small baryon asymmetry left matter after annihilation. The exact cause is still studied.

### 12.4 Explain pair annihilation with an example. Explain its utility in medicine.

An electron and positron annihilate into two 0.51 MeV gamma photons. In PET scans, positron-emitting tracers produce annihilation photons that are detected to image body processes.

### 12.5 Explain conservation of energy and momentum in electron–positron pair annihilation.

The total rest energy 1.02 MeV becomes two photons of 0.51 MeV each, emitted in opposite directions so total momentum remains zero.

### 12.6 Describe protons and neutrons in terms of quark composition.

Proton = uud, neutron = udd. The up quark has charge +2/3 e and down quark −1/3 e, giving proton +1 e and neutron 0.

### 12.7 Describe the four fundamental forces in nature.

Strong: binds quarks and nuclei. Electromagnetic: acts between charges. Weak: responsible for beta decay. Gravity: acts between masses, weakest at particle scale.

### 12.8 Describe the classification of elementary particles.

Particles are classified as quarks, leptons and exchange bosons. Hadrons made of quarks include baryons (three quarks) and mesons (quark–antiquark). Leptons include electrons and neutrinos. Bosons carry forces.

## Numerical Problems (fully worked)

### 12.1 Uranium-238 is an alpha emitter. Find A, Z and symbol of daughter.

**Given:** parent: A = 238, Z = 92

**Formula:** alpha decay: A − 4, Z − 2

A = 234, Z = 90, element thorium (Th).

**Answer: A = 234, Z = 90, Th**

### 12.2 Polonium-218 is a beta-minus emitter. Find A and Z of daughter.

**Given:** parent: A = 218, Z = 84

**Formula:** $\beta^-$: A unchanged, Z + 1

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A = 218, Z = 85 (astatine).

**Answer: A = 218, Z = 85**

### 12.3 Nitrogen-14 bombarded by alpha gives oxygen-17. What is the product particle?

**Given:** $^{14}N + {}^{4}He \rightarrow {}^{17}O + X$

**Formula:** conserve A and Z

A: 14 + 4 = 17 + A_X → A_X = 1. Z: 7 + 2 = 8 + Z_X → Z_X = 1. X is a proton $^{1}H$.

**Answer: proton ($^{1}H$)**

### 12.4 Show nucleon number and charge number are conserved in 12.3.

Nucleons: 14 + 4 = 17 + 1 = 18. Charge: 7 + 2 = 8 + 1 = 9. Both conserved.

**Answer: Conserved**

### 12.5 Determine rest-mass energy of electron in eV if rest mass is 0.000555 u.

**Given:** m = 0.000555 u, 1 u ≈ 931 MeV

**Formula:** $E = mc^2$

E = 0.000555 × 931 ≈ 0.517 MeV.

**Answer: 0.517 MeV**

Accepted electron mass 0.000549 u gives 0.511 MeV; the book uses 0.000555 u.

### 12.6 Calculate Q-value for Rutherford's reaction: $^{14}N + {}^{4}He \rightarrow {}^{17}O + {}^{1}H$. Masses: 14.007515 u, 4.003837 u, 17.004533 u, 1.008142 u.

**Given:** masses as printed

**Formula:** $Q = (m_{initial} - m_{final})c^2$

Initial = 18.011352 u. Final = 18.012675 u. Δm = −0.001323 u.

Q = −0.001323 × 931 ≈ −1.23 MeV.

**Answer: −1.23 MeV**

## Textbook checks flagged in this chapter

• Numerical 12.5: book uses electron mass 0.000555 u, giving 0.517 MeV; accepted value gives 0.511 MeV.

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